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Assignment 10: Explainable artificial intelligence (XAI)

What would you advocate for bringing transparency or explainability in an ML/AI system? What are the advantages to the users as well as the organization providing that system?

I would like to advocate for bias and fairness reporting in an ML/AI system. You would want a system to be free of bias to allow human oversight in data-driven decisions. An example will be a situation in healthcare when obtaining data and applying the ML system to output information free of bias such as race, ethnicity, or "high risk" outcomes. The advantages of applying ML system for users free of bias and is fair builds trust and confidence. It allows users to feel safe by knowing how and why decisions are made. The advantages for organizations help developers and stakeholders gain insights of model behavior with well understood applications to understand outcomes by translating technical reasoning into plain language. Advocating bias and fairness in a ML system brings transparency and can be used as tools users can rely on and bridge the gap between complex algorithms and human understanding.

What is a "fair" system? Take an example of an ML/AI system and tell us what achieving "fairness" would look like.

A "fair" system is system that can be designed and implemented to avoid discriminatory decisions by allowing equitable treatment and outcomes for different individuals and groups (Arrieta et al.,2020). An example of an ML/AI system could be a logistic regression model to predict the likelihood of an individual getting a loan from a bank. To achieve fairness in this model is to be aware of sensitive attributes such as race or ethnicity in the pre-processing step of the model. Another is to see if the attribute(s) is not being directly used as an input feature for the prediction model using SHAP to identify other attributes in the data that can influence predictions for certain racial groups (Arrieta et al., 2020). Another way to achieve fairness is using a technique called reweighing to adjust the weights of the training data points to reduce the impact of proxy features. Using these techniques can help the ML model to make decisions that are not influenced by an individual's race or where they live to prevent discriminatory outcomes in this situation.

How do you see concepts like bias, fairness, accountability, and transparency-related?

I see these concepts as a framework to make accurate data-driven decisions to improve organizations and help them finalize a logical outcome to stay compliant in situations like healthcare, finance, or hiring. I think transparency is important when obtaining data because it builds trust for users such as data scientists to critically analyze data and interpret how the model used made these outcomes. Having data free of bias and fairness reporting is also important because it builds accountability to accurately publish results and builds credibility. Applying these fundamentals prevents discrimination, legal risk, and fosters loyalty to the organization for their customers.

Give an example of a system that is biased but fair.

An example where a system is biased but fair can be approval for mortgage to home buyers. Implementing an AI system to determine whether the individual can be approved or not for a mortgage can be biased such as socio-economic status, personal income, credit score, or debt. The system can favor individuals who are in an upper socio-economic status than low-income groups or favor first-time home buyers. However, the system can be fair by correcting features of bias (minorities or low-income families) and allowing transparency for applicants to know what criteria is factored in to get approval for mortgage. These build trust for both lenders and buyers to allow equitable access to mortgages and compliance for home buyer organizations to follow.

Reference

Arrieta, B.A., Diaz-Rodriguez, N., Del Ser, J. et al. (2020). Explainable artificial intelligence (XAI): Concepts, taxonomies, opportunities, and challenges towards responsible AI. *Information Fusion, 58,* 82-115. https://doi.org/10.1016/j.inffus.2019.12.012